

Drew Berry

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A strong technical programmer with first-class academic credentials and a passion for AI research. During my postgraduate studies, I presented my thesis to a research group and received high commendation, with a game engine I developed later adopted by another student for their own research. I've also collaborated with the University of Essex and Oracle to deploy a large language model on Oracle Cloud Infrastructure, and I am currently expanding my MLOps skills through a local pipeline simulation project.

My university professor wrote, "***Drew Berry was easily in the top 5% in his cohort at software engineering. He shows an independent ability and determination to solve problems reliably. I would therefore heartily recommend him as a really good hiring choice for any company.***" Reference available.

SKILLS

Programming: Python (6 years), Java (4 years), SQL, Minimal Web Dev (HTML, CSS, JS)

Frameworks/Libraries: NumPy, TensorFlow/Keras, PyTorch, Pandas, Seaborn, scikit-learn, Matplotlib, LangChain, SQLite3, Streamlit, transformers, FastAPI

Tools: JetBrains suite, Git/GitHub, Jira, Google Colab, Jupyter Notebook, Oracle Cloud (OCI), Hugging Face

- Generative AI and LLMs
- Retrieval-Augmented Generation (RAG)
- Neural Network Design
- Data Cleaning and Preprocessing
- Technical Documentation
- MLOps
- Natural Language Processing (NLP)
- Reinforcement Learning
- Advanced Machine Learning Techniques
- Database Management

EDUCATION

MSc Artificial Intelligence | **Distinction** | University of Essex Oct 2023 - Oct 2024

Relevant Modules: Machine Learning, Neural Networks and Deep Learning, Intelligent Systems and Robotics, Professional Practice and Research Methodology, Text Analytics, Game Artificial Intelligence

BSc Computer Science | **Honours Class I** | University of Essex Oct 2020 - Jul 2023

Relevant Modules: Evolutionary Computation and Genetic Programming, Languages and Compilers, Network Security, Algorithmic Game Theory, Computer Security, Data Structures and Algorithms, Computer Game Programming, Software Engineering

PROJECTS

RAG System for PDF Question Answering | Python | Self-Directed Project | [GitHub Link](#)

- Deployed a LLaMA 3.2 1B model on *Oracle Cloud Infrastructure* (OCI), via a cloud notebook, to experiment with generative AI and Retrieval-Augmented Generation (RAG) workflows.
- Developed a custom HuggingFace-compatible embedding class and integrated *LangChain* orchestration to build a PDF document question-answering system, improving contextual response accuracy through advanced text preprocessing with NLTK.
- Completed in collaboration with the University of Essex and Oracle.

AI-powered Music Recommendations App | Python | Personal Project | [GitHub Link](#)

- Designed and implemented an AI-powered web app using *Streamlit* that delivers personalised music recommendations from natural language prompts.
- Integrated a *large language model* with *Last.fm API* metadata to interpret user intent and generate nuanced, vibe-based track suggestions beyond simple genre or mood filters.

MLOps Pipeline Simulation | Python | Personal Project | [GitHub Link](#)

- Designed and implemented a fully local, cloud-inspired MLOps pipeline to demonstrate end-to-end machine learning system design without relying on paid cloud services.
- Built modular components simulating object storage, data warehousing, model registries, orchestration, and API gateways using tools such as FastAPI, Streamlit, TensorFlow, and SQLite3.
- Showcased key MLOps principles including reproducibility, automation, modularity, and lifecycle versioning through a reusable, task-agnostic pipeline architecture.

MSc Thesis | Python | University Project | [GitHub Link](#)

- Developed AI driving models for the TORCS racing simulation, comparing the performance of genetic and PPO reinforcement learning algorithms.
- Tuned hyperparameters and reward functions for optimal results, eliciting complex driving behaviours.
- Fixed bugs in a *community-written* game client.
- Presented findings to both an assessor and a research group, receiving high commendation.

Kaggle Competitions | Python | Personal Project | [Kaggle Link](#) | [GitHub Link](#)

- Actively participating in Kaggle competitions to sharpen my data science and machine learning skills, applying advanced techniques to real-world problems and achieving strong results in various challenges.

CERTIFICATIONS

Databases and SQL for Data Science with Python | IBM | [View PDF](#)

Dec 2024

- Gained expertise in querying, filtering, and aggregating data using SQL to extract insights from large datasets.
- Learned to use the *SQLite3* library for database management, manipulation, and visualisation in Jupyter Notebooks.
- Completed an honours module on advanced SQL techniques, including stored procedures, transactions, and joins, to handle complex database operations efficiently.

Generative AI | IBM | [View PDF](#)

Jan 2025

- Gained expertise in using generative AI for automating ETL workflows, schema design, data pipelines, repository maintenance, and database querying, with hands-on labs and a focus on *responsible AI practices* and real-world applications.

EMPLOYMENT HISTORY

Concession Supervisor | Menkind Limited

Sept 2020 - Jan 2021

- Assisted customers with merchandise inquiries and selections, ensuring a positive experience.
- Applied product knowledge to provide accurate information.
- Restocked and maintained merchandise displays for a clean and organised presentation.
- Worked to a series of KPIs, including upselling, display standards and stock control.

Farmhand | DeBohuns Hall Farm

Weekends 2017 - 2020

- Helped in every area of the farm, all year round.
- This includes basic equipment operation, animal handling, maintenance, basic carpentry and construction, as well as the use of manual and mechanical tools.

REFERENCES

Excellent academic references and professional endorsements from prior collaborators are available upon request.